# Fenghao **Liu**

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## Education

**University of Michigan** — *Ann Arbor, MI, USA*

M.S. in Robotics | **GPA:4.0/4.0** — *Aug. 2025 - May. 2027*

**Southeast University** — *Nanjing, China*

B.S. in Information and Computing Science | **GPA:3.53/4.0** — *Sep. 2021 - Jun. 2025*

## Experience

**Real-Time Neural Rendering & 3D Reconstruction (Accepted to OCEANS 2026)** — *Ann Arbor, MI, USA*

Research Assistant | FROG Lab, University of Michigan — *Mar. 2026 - Present*

- **Implicit 3D Perception with Sparse Sensors:** Addressed the extreme sparsity and ill-posed nature of side-scan sonar data by building a baseline for implicit 3D perception from scratch based on the SIREN architecture and Lambertian reflectance model; evaluated NeuSSS and SIREN implicit neural representation models on real-world datasets.
- **Physical Model Tuning & Evaluation:** Analyzed underwater acoustic reflection characteristics to refine networks and physical rendering models through ablation studies; extracted 3D scene representations on simulated and real-world underwater datasets, providing a reliable benchmark for underwater robot spatial perception.

**Tuniu (Xiao Niu Travel Assistant)** — *Nanjing, China*

AI Agent Algorithm Intern — *May 2025 - Aug. 2025*

- **Multi-Agent Routing:** Designed a primary dispatch layer with Qwen3 and async verification mechanisms, achieving 93% routing correctness and an average latency of 2.57s.
- **Retrieval Optimization:** Deployed Milvus vector database and fine-tuned Sentence Transformer models, boosting retrieval precision by 2%.
- **Automated Evaluation System:** Developed automated factuality checks for POI data and itinerary feasibility, fixing critical data issues and enhancing the robustness of the production pipeline.

## Selected Projects

**Closed-Loop Deployment & Offline DPO for Autoregressive VLMs (g0.5) on Real Manipulators** — *Ann Arbor, MI, USA*

Graduate Independent Project | University of Michigan — *Jun. 2026 - Present*

- **End-to-End Pipeline & SFT Baseline:** Calibrated the SO101 leader-follower manipulator and built a dual-camera teleoperation environment using LeRobot; collected 156 expert demonstrations to perform full SFT (4 epochs) on the g0.5 foundation model, enabling basic cross-task generalization (e.g., placing toilet paper/blocks).
- **Offline Trajectory Preference Construction:** Addressed SFT failures in edge cases by collecting 56 policy rollouts and interventions (success/failure trajectories) across 8 edge initialization buckets; designed an offline preference data pipeline pairing successful and failed trajectories within the same task and bucket, unrolled into fixed-length sequence windows to construct high-quality contrastive datasets.
- **Autoregressive Action-Space DPO Implementation:** Refactored the TPO paradigm from the GRAPE framework into the training stack; calculated Reference and Policy log-probabilities precisely at the g0.5 autoregressive action-token level, designing a joint loss function (DPO Loss + Chosen CE). Mitigated OOM issues during full fine-tuning via offline pre-computation and caching of Reference logprobs, achieving significantly better convergence stability than pure SFT in just 1 epoch.
- **Action Representation & Closed-Loop Deployment:** Reconstructed action representations for the SO101 joint space; built a low-latency remote inference service via WebSocket + SSH tunnel, integrated with action clipping and replanning mechanisms to achieve highly robust closed-loop interaction in complex physical environments.

**Fine-Tuning & Ablation Analysis of Diffusion VLA Models ($\pi_0$)** — *Ann Arbor, MI, USA*

Graduate Independent Project | University of Michigan — *Jun. 2026*

- **Pipeline Reproduction & Efficient Fine-Tuning:** Reproduced the $\pi_{0.5}$ underlying training pipeline on 4x A100 (40GB) GPUs using the OpenPI/JAX/Flax framework, conducting large-scale multi-modal fine-tuning experiments on the LIBERO-Spatial simulation dataset.
- **Modular Fine-Tuning & Ablation Analysis:** Compared the performance of full fine-tuning, PaliGemma LoRA, and Action Expert LoRA; quantitatively analyzed the root cause of suboptimal performance in pure action-expert fine-tuning, validating that the spatial reasoning capability of VLA models is highly coupled with the underlying logic of vision-language foundation models (e.g., SigLIP/PaliGemma), providing crucial theoretical support for real-world model selection.

**Loop Closure System for Deep SLAM (DROID-SLAM)** — *Ann Arbor, MI, USA*

Graduate Project | University of Michigan — *Feb. 2026 - Apr. 2026*

- **Architecture Design:** Designing a dual-branch visual place recognition (VPR) pipeline fusing CLIP semantic embeddings with structural line features (e.g., DeepLSD) to overcome low-texture and illumination challenges.
- **Backend Optimization:** Integrated multi-modal line constraints into the DROID-SLAM factor graph to enable robust global pose optimization, utilizing an adaptive robust kernel truncation mechanism to correct monocular scale drift and reduce ATE errors on long sequences like KITTI.

**Control System for Autonomous Mobile Robot & 5-DoF Manipulator** *Ann Arbor, MI, USA*

GRADUATE PROJECT | UNIVERSITY OF MICHIGAN *Aug. 2025 - Dec. 2025*

- **Navigation & Low-Level Control:** Implemented a complete SLAM system from scratch on the MBot embedded platform using Particle Filter (MCL) and ray-tracing models; developed path planning and autonomous exploration strategies, deploying a hybrid Feedforward + PID controller to overcome nonlinear friction, and integrated AprilTag visual relocalization for precise pallet manipulation.
- **Kinematics & Visual Calibration:** Developed high-precision Forward Kinematics via Product of Exponentials (PoX) (avg. error 1.9%) and a Geometric Inverse Kinematics (IK) solver; applied AprilTag PnP and Homography transformations for automated extrinsic calibration and perspective distortion correction, building a visual-guided "Click-to-Grab" sorting system.

## Skills

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| **Programming** | C++, Python, Linux, ROS 2, Docker, Git, CMake, OpenCV, Hugging Face |
| **Robotics** | SLAM, 3D Reconstruction, Teleoperation, Motion Planning (A*, RRT), Kinematics & Dynamics, PID Control |
| **AI & Data** | PyTorch, JAX/Flax, NeRF, 3DGS, Vision-Language Models, VLA, LoRA, LeRobot, OpenPI |